Newton's forward & backward interpolation

$y = f(x)$

$x : x_0 \quad x_1 \quad x_2 \quad x_3 \; \cdots \; x_n$

$y : y_0 \quad y_1 \quad y_2 \quad y_3 \; \cdots \; y_n$

$h = x_1 - x_0 = x_2 - x_1 = x_3 - x_2 \cdots$

$y = ?$ at 'x' which lies near to x_0

The forward interpolation formulae is given by

$$\boxed{y = y_0 + P(\Delta y_0) + \frac{P(P-1)}{2!}\Delta^2 y_0 + \frac{P(P-1)(P-2)}{3!}\Delta^3 y_0 + \frac{P(P-1)(P-2)(P-3)}{4!}\Delta^4 y_0 \cdots \frac{P(P-1)(P-2)\cdots[P-(n-1)]}{n!}\Delta^n y_0}$$

$P = \frac{x - x_0}{h}$

Backward interpolation formuale.

$y = ?$ at x which lies near to x_n

$$y = y_n + r(\nabla y_n) + \frac{r(r+1)}{2!}\nabla^2 y_n + \frac{r(r+1)(r+2)}{3!}\nabla^3 y_n + \frac{r(r+1)(r+2)(r+3)}{4!}\nabla^4 y_n \cdots + \frac{r(r+1)(r+2)(r+3 \cdots r+(n-1))}{n!}\nabla^n y_n$$

where $r = \frac{x - x_n}{h}$

① Given $f(0)=1$ $f(1)=3$ $f(2)=7$ $f(3)=13$ find $f(0.1)$ using Newton's interpolation.

$\Rightarrow$ $x : 0 \quad 1 \quad 2 \quad 3$

$y : 1 \quad 3 \quad 7 \quad 13$

$y = ?$ at $x = 0.1$ which lies near to x_0.
$\therefore$ Newton forward interpolation is appropriate.

Newton forward formula is given by.

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0$$
$$+ \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 y_0 \quad \text{--- (1)}$$

Forward difference table

x	y	Δy	$\Delta^2 y_0$	$\Delta^3 y_0$
0	1	$2 = \Delta y_0$	$2 = \Delta^2 y_0$	0
1	3	4	2	
2	7	6		
3	13			

$y_0 = 1 \quad \Delta y_0 = 2 \quad \Delta^2 y_0 = 2$

eq^n (1) becomes

$$y = 1 + (0.1 \times 2) + \frac{(0.1)(0.1-1)(2)}{2} + \frac{(0.1)(0.1-1)(0.1-2)(0)}{6}$$

$y = 1 + 0.2 + 0.1 \times (0.9)$
$y = 1.1100$

$$p = \frac{x - x_0}{h} \quad = h = x_1 - x_0 = \frac{0.1}{1} = 0.1$$

$$y = 1 + (0.1) \times 2) + \frac{(0.1)(0.1-1)2}{2!} + 0$$

$y = \underline{\underline{1.11}}$

(2) Given

	x_0	x_1	x_2	x_3	x_4	x_5
x :	1	1.2	1.4	1.6	1.8	2
y :	0	0.128	0.544	1.296	2.432	4.00
	y_0	y_1	y_2	y_3	y_4	y_5

Find (1.1)

=> y = ? at f(1.1) which lies near to x_0

The Newton's forward interpolation is appropriate.

∴ Newton forward intepolation is given by

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 y_0$$

where,

$$p = \frac{x - x_0}{h} = \frac{1.1-1}{0.2} = 0.5$$

Forward difference table

x	y	Δy	$\Delta^2 y_0$	$\Delta^3 y_0$	$\Delta^4 y_0$	
1	0	0.128	0.288	0.048	0	0
1.2	0.128	0.416	0.336	0.048	0	
1.4	0.544	0.752	0.384	0.048		
1.6	1.296	1.136	0.432			
1.8	2.432	1.568				
2	4.00					

$y_0 = 0$ $\Delta y_0 = 0.128$ $\Delta^2 y_0$ 0.288 $\Delta^3 y_0 = 0.048$

$$y = 0 + 0.5(0.128) + \frac{(0.5)(0.5-1)}{2}(0.288) + \frac{(0.5)(0.5-1)(0.5-2)}{6}(0.048) + 0$$

$y = 0.031$

(3) The area of a circle 'A' Corresponding to diameter D is given below.

D :	80	85	90	95	100
A :	5026	5674	6362	7088	7854

find the area when D is 105 units using appropriate interpolation

=> A = ? at D = 105 units which lies near D_n So Newton's Backward interpolation.

Formula.

$$y = y_n + r\nabla y_n + \frac{r(r+1)}{2!}\nabla^2 y_n + \frac{r(r+1)(r+2)}{3!}\nabla^3 y_n + \frac{r(r+1)(r+2)(r+3)}{4!}\nabla^4 y_n$$

Backward difference

D	A	∇A	∇A^2	∇A^3	∇A^4
80	5026	648	40	-2	4
85	5674	688	38	2	
90	6362	726	40		
95	7088	766			
100	7864				

$y_n = 7854$. $\nabla y_n = 766$ $\nabla^2 y_n = 40$ $\nabla^3 y_n = 2$

$\nabla^4 . y_n = 4$

$$r = \frac{x - x_n}{h} = \frac{105 - 100}{5} = 1$$

$$y = 7854 + 1(766) + \frac{1(1+1)40}{2} + \frac{(1(1+1)(1+2)2)}{6}$$

$$+ \frac{1(1+1)(1+2)(1+3)(4)}{24}$$

$\boxed{y = 8666}$

4) Given

$\sin(45^\circ) = 0.7071$ $\sin(50^\circ) = 0.7660$

$\sin(55^\circ) = 0.8192$ $\sin(60^\circ) = 0.8660$

Find $\sin(42^\circ)$ & $\sin(57^\circ)$ using interpolation

$\Rightarrow$

x:	$\sin(45^\circ)$	$\sin(50^\circ)$	$\sin(55^\circ)$	$\sin(60^\circ)$
y:	0.7071	0.7660	0.8192	0.8660

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
45	0.7071			
		0.0589		
50	0.7660		-5.7×10^{-3}	
		0.0532		-7×10^{-4}
55	0.8192		-6.4×10^{-3}	
		0.0468		
60	0.8660			

$$p = \frac{x - x_0}{h} = \frac{42 - 45}{5} = -0.6$$

<u>Forward interpolation</u>

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 y_0$$

$$= 0.7071 + (-0.6)0.0589 + \frac{(-0.6)(-0.6-1)}{2}(-5.7\times10^{-3}) + \frac{(-0.6)(-0.6-1)(-0.6-2)}{3!}(-7\times10^{-4})$$

$$\boxed{y = 0.66932}$$

<u>Backward interpolation</u>

$$y = y_n + r\nabla y_n + \frac{r(r+1)}{2!}\nabla^2 y_n + \frac{r(r+1)(r+2)}{3!}\nabla^3 y_n$$

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$$+\frac{r(r+1)(r+2)(r+3)}{4!}\nabla^4 y_n$$

$$= 0.8660 + (-0.6\times 0.0468) + (-0.6)(-0.6+1)\times(-6.4\times10^{-3})$$
$$+\frac{(-0.6)(-0.6+1)(-0.6+2)}{6}(-7\times10^{-4})$$

$$\boxed{y = 0.83873}$$

★ The population of town is given by the following data.

year :	1961	1971	1981	1991	2001
Population : (Thousand)	19.96	39.65	58.81	77.18	94.58

Calculate increase in the population from the the year (1965 to 1995).

Lagranges interpolation and inverse interpolation.

$y = f(x)$ $y = ?$ at $x = '$

x : x_0 x_1 x_2 .... x_n
y : y_0 y_1 y_2 .... y_n

$$y = \frac{(x-x_1)(x-x_2)(x-x_3)\cdots(x-x_n)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)\cdots(x_0-x_n)}y_0$$

$$+\frac{(x-x_0)(x-x_2)(x-x_3)\cdots(x-x_n)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)\cdots(x_1-x_n)}y_1$$

$$+\frac{(x-x_0)(x-x_1)(x-x_3)\cdots(x-x_n)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)\cdots(x_2-x_0)}y_2$$

$\vdots$

$$+\frac{(x-x_0)(x-x_1)(x-x_2)\cdots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)(x_n-x_2)\cdots(x_n-x_{n-1})}(y_n)$$